quantfit

Making a big AI model fit on the computer we already own.

Milestone 1 review — July 27, 2026

The problem

- Powerful AI models are big. The one we want is about **98 GB**.
- Our video card has **24 GB** of memory. The model does not fit.
- You can "shrink" a model to save space, but shrink it too much and it gets noticeably dumber.
- Shrinking everything the same amount does not work here: the size that fits ruins the model, and the size that keeps it smart doesn't fit.

The idea

- Not all parts of a model are equally fragile.
- Some parts can be squeezed hard with no real harm. Others fall apart if you touch them.
- **quantfit measures which parts are which**, then squeezes each part just the right amount to fit our card.

Before → After

Before this milestone

- The idea existed only as documents.
- "Will it even fit?" was a guess based on rough estimates.

After this milestone

- A working planning tool you can run today.
- It reads the real model's blueprint and tells us exactly how much room we have — no more guessing.
- Given measurements, it builds the full "shrink plan" automatically, and can explain every choice it made.

What we built

- A **budget calculator**: how much memory is really free for the model after everything else takes its share.
- A **recipe maker**: decides how much to squeeze each part, staying inside the budget, and says why for every decision.
- If a plan is impossible, it says so honestly — and by how much.
- 88 automated tests check all of it on every change.

What the numbers told us

- With the real model's blueprint: about **20 GB** is truly available for the model on our 24 GB card.
- That means squeezing to about **3.5 units of size** on average — the serving software we planned to use only supports 4.
- So there's a known gap to close. We wrote down four ways to close it and will pick one in the next phase.
- Finding this out **now**, on paper, cost nothing. Finding it after months of building would have hurt.

What's next

- Build the **scanner**: the tool that measures which parts of the model are fragile. This is the science step.
- Then use those measurements with today's recipe maker to produce the first real shrunken model.
- Progress and decisions are tracked in the project repo: `github.com/Alberto-Codes/quantfit` (pull request #1).